

Ali Sufiyan Khan

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🌐 LinkedIn | 🐙 GitHub | 🌐 Portfolio

Skills

Cloud & DevOps: Kubernetes, Docker, AWS (EC2), GCP, Linux (Bash), Tailscale, Cloudflare Tunnels, Portainer, Netdata

Languages: Python, C, Java, SQL

AI/ML & Frameworks: PyTorch, TensorFlow, NVIDIA Isaac Lab, NVIDIA Isaac Sim, Stable Baselines3, Gymnasium, LangChain, Scikit-learn, OpenCV

Developer Tools: Git, GitHub, Cursor, Antigravity, OpenAI Codex, Claude Code, Hermes Agent, JetBrains IDEs

Education

Vellore Institute of Technology (VIT Vellore), B.Tech in Computer Science and Engineering Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence

Projects

Deep Reinforcement Learning for 6DOF Robotic Arm Pick-and-Place Control



Python | PyTorch | NVIDIA Isaac Lab | Isaac Sim | PPO

- Developing a UR10e 6DOF pick-and-place system with a Robotiq 2F-85 gripper using NVIDIA Isaac Lab and Isaac Sim.
- Implemented PPO training for staged reaching, grasping, lifting, and placement; completed **Phase 0 reach training**.
- Built vectorized environments with custom observations, reward shaping, collision handling, and curriculum progression.

F1 Reinforcement Learning Agent



Python | Stable Baselines3 | Gymnasium | UDP Telemetry | vgamepad

- Developed an autonomous racing agent for F1 22 using Python, Stable Baselines3, and PPO on game telemetry.
- Built a **60 Hz** UDP telemetry parser and a custom Gymnasium environment with virtual-LiDAR track-deviation features.
- Implemented continuous steering, throttle, and braking through a virtual Xbox controller using vgamepad.

Hybrid Multi-Cloud Kubernetes Orchestration Platform

K3s | Docker | AWS | GCP | Tailscale | Cloudflare Tunnels | Locust

- Designed a hybrid cluster integrating on-premise edge, AWS, and GCP nodes using K3s and Tailscale.
- Implemented cloud bursting and secured public service exposure through Cloudflare Tunnels.
- Sustained **1,000+ concurrent users** with a **0% failure rate** in distributed Locust load testing.

RoleAtlas – AI-Assisted Global Job Discovery Platform



Next.js | TypeScript | Rust | NATS JetStream | PostgreSQL | Docker

- Building a qualification-first workspace for discovering, ranking, saving, and tracking global internships and entry-level roles.
- Developed resume-based suitability scoring and AI-assisted application dossiers with traceable evidence, constraint checks, and truthful tailoring.
- Engineered a distributed Rust crawler with durable queues, retries, deduplication, responsible crawling controls, and structured ATS ingestion.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Aug 2025

- Issued by Oracle University | Valid through August 2027